

**LearnQuest: An Adaptive Learning Management System
Powered by Artificial Intelligence for STEM Students of
Kapitolyo High School**

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CHAPTER 1

INTRODUCTION

The Problem and its Background

Students in the field of Science, Technology, Engineering, and Mathematics (STEM) commonly find it difficult to keep their engagement, motivation, and consistency needed to succeed academically, especially in their science subjects. Students find it challenging to continue their interests and efficiently manage their productivity because the science subjects' content contains a lot of information and requires frequent analysis. Additionally, despite providing access to educational tools and standard instructional resources of traditional and online learning platforms, they are still unable to address the unique needs of the students.

In response to these learning challenges of the students, educational institutions are increasingly adopting innovative digital learning tools that could blend technology and the active participation of the students. Specifically, the embedding of artificial intelligence (AI) with learning management systems (LMS) has proven to be effective for students in creating more adaptive learning.

The advancements in digital technology have significantly reshaped the method of delivery and management of education. As stated by Hashemi-Pour et al. (2024), the utilization of Learning Management Systems (LMS) is one of the most important content management tools that teachers can easily use to organize and manage class lessons, communication, and test creation. Additionally,

Prahani et al. (2022) emphasized that the LMSs provide centralized control of learning materials and analytics by improving teaching efficiency and student participation.

Learning Management System (LMS) has become very important for convenient learning, progress monitoring, and effective teaching and learning by means of the education process. According to Lee D. (2023), the importance of implementing LMS in the secondary education sector is that the educational institutions that are making use of LMS can implement flexible learning and engaging means of education before and after the COVID-19 pandemic.

Therefore, there is an indication of the rise in usage of LMS through platforms that provide advanced features like flexible educational approaches, real-time assessments, analytics support, and Artificial Intelligence (AI). As Aljad et al. (2025) observe, this development is the result of the fast technological development of LMS platforms.

Despite advancements in digital technology, uses of online education continues to encounter many problems. Most students, especially senior high school students, find it difficult to concentrate while learning online. While Prahani et al. (2022) note that although Learning Management Systems (LMS) are extremely efficient in managing lessons and tests, students lack motivation due to the absence of personal interaction. The usage of artificial intelligence (AI) would be helpful as a solution to these problems because it provides personalized feedback and adaptive learning.

The significant contribution of Artificial Intelligence (AI) within the academic field seems to be growing tremendously because of its ability to observe student performance and learning processes. It is also capable of providing customized material according to the specific needs of students. AI technology is a very effective way to enhance the process of teaching and learning by integrating it into an LMS system.

Kapitolyo High School (KHS) was established in 2004 on 2nd of June, and was located on Barangay Kapitolyo, San Ignacio St., Pasig City. Since then, this institution has consistently worked hard to improve its instructional quality to provide an outstanding educational experience for its students. Currently, KHS has a total of approximately 1,700 senior high school students, comprising 844 Grade 11 students and 856 Grade 12 students, and a total of 40 faculty teachers who provide support to the students.

In addition, KHS provides students with three main options for their studies: Science, Technology, Engineering, and Mathematics (STEM), Accountancy, Business, and Management (ABM), and Humanities and Social Science (HUMSS). In addition to those main options, there are also Technical–Vocational–Livelihood (TVL) courses such as Animation, Computer Systems Servicing, and Home Economics. Hence, these diverse tracks enable students to develop their specialized skills while exploring their career interests.

Google Classroom was represented as one of the main Learning Management Systems (LMS) used at Kapitolyo High School during the COVID-19

pandemic. Due to the suspension of in-person classes, the application proved to be a perfect aid tool for teachers to offer lessons, assign homework, and assess their students, without which the education process wouldn't have taken place. Particularly, Kapitolyo High School is still utilizing Google Classroom in the current day-to-day teaching process.

Nonetheless, despite the successful use of Google Classroom for learning delivery has faced several issues that impact the quality of the learning process. The application acts as a repository for lessons and grade management, but lacks features that can increase the engagement of students and personalization. Moreover, Google Classroom lacks the advantages of AI, which means that adaptive learning analytics and smart feedback cannot be produced in the given platform.

These gaps highlighted the necessity of developing a smarter and more adaptive learning management system that would be able to assist STEM students at Kapitolyo High School in their science subjects. Addressing these problems is possible by means of using a system that will increase active engagement, provide interactive personal assistance, and help with continuous improvement. Integrating artificial intelligence into the learning management system can make it smarter and a more learner-focused environment.

This study aims to develop a system, LearnQuest, an adaptive learning management system powered by Artificial Intelligence for Grade 11 and 12 STEM Students of Kapitolyo High School. Artificial Intelligence (AI) seeks to design and

develop a user-centered frontend architecture that includes an interactive quiz model for AI automatic generation of the questions on student assessment, summarization of the science learning modules, and provides personalized feedback that is based on the uploaded PDF and PowerPoint materials, and provides a personalized student dashboard that shows performance analytics, mastery, and AI study suggestions.

To power these core intelligent functionalities, the system will integrate a Large Language Model (LLM), specifically Llama 3.3-70B-Instruct. This model will allow for automatic creation of assessment questions directly from uploaded learning material, intelligent summarization of subject science modules into understandable content, and personalized feedback generation in accordance with students' performance.

Alongside the LLM, LearnQuest will employ Bayesian Knowledge Tracing (BKT) as the core adaptive learning algorithm, which will be able to assess and predict students' mastery and pinpoint the hidden learning gaps in their learning and progress in the STEM field. In this way, the adaptive learning experience will be able to understand what needs to be reinforced by the student and when the student can move to the next level.

For educators, the system provides a comprehensive learning analytics dashboard that will be created with graphical illustrations, such as radar charts, which will display student mastery levels in various competencies. This dashboard

delivers real-time analytics on assessment performance, completion rate, and BKT-based mastery probability.

The proposed system would help students improve their motivation and engagement in learning through enjoyment. The capacity of the system to analyze the strengths and weaknesses of the students through artificial intelligence will help the students enhance their performance in science subjects. The system will also assist the teachers because it provides them with information on how the students are performing in real time. In addition, the school administration could use this system to help manage learning as well as collect measured data about their development.

This study will emphasize the importance of technology in today's education, as it will show that integrating a Learning Management System (LMS) and artificial intelligence (AI) could effectively enhance and assist learning experiences. Through monitoring engagement challenges and providing adaptive assistance, LearnQuest aims to improve the learning experience of STEM students in the science subject at Kapitolyo High School.

Review of Related Literature

This chapter presents a review of existing literature and researchers that constitute the theoretical background for designing LearnQuest: An Adaptive Learning Management System Powered by Artificial Intelligence. The purpose of

this chapter is to give an in-depth analysis of existing technologies in the field of adaptive learning and artificial intelligence.

Learning Management System (LMS) and Evolving Role in Education

The learning management system (LMS) has become an essential part of modern education due to its capabilities in managing learning materials, facilitating communication, monitoring student progress, and overseeing educational tasks. The fast growth of educational technologies brought about the fact that the LMS evolved from being simply a digital repository of learning materials to becoming an intelligent, interactive learning environment.

One of the major strengths of modern learning management systems is their ability to promote learner engagement and accessibility. As Bonilla-Priego (2024) emphasized, the real success of the effectiveness of any LMS depends not only on its capacity to offer digital learning materials but also on its ability to increase availability and strengthen the interactions between teachers and students. A well-designed LMS facilitates the creation of an interactive environment where educators can seamlessly control their students' work. This kind of technological evolution of the platform indicates the increasing need for digital learning environments that would actively support adaptive educational and instructional purposes.

Beyond delivering content, modern LMS platforms generate valuable learner data that teachers can utilize to evaluate academic performance and improve their teaching methods. As mentioned by İskigöz (2024), the students' interaction monitoring, such as the quiz attempts, duration spent learning, and login frequency, can effectively be used to predict academic outcomes of the students. These learning analytics generated through a learning management system (LMS) can provide valuable information about students' learning habits that can assist teachers in pinpointing the problems in learning and implementing the targeted interventions for such problems.

Furthermore, the integration of learning analytics in the learning management systems has enabled the utilization of artificial intelligence (AI) in education. Through constant monitoring and analysis of learner interactions, artificial intelligence technology can detect learning trends, predict student performance, and suggest relevant learning tasks to learners based on their personal needs. Hence, modern platforms' LMS have become intelligent learning environments that support truly adaptive instruction and improve academic performance.

This body of literature illustrates that learning management systems (LMS) can act as the technological platform for developing intelligent educational solutions because the LMS platforms organize learning materials, gather data on students, and perform learning analytics, making them good platforms for incorporating intelligent learning solutions for an adaptive learning framework. All

these features make it directly relevant to the proposed LearnQuest system, which will use an LMS with features that are integrated with artificial intelligence to track and measure the competencies of Grade 11 and Grade 12 STEM learners.

However, Learning Management Systems (LMS) provide all the required technical infrastructure for digital learning. Modern studies in education indicate that simply delivering the material is no longer enough to meet the diverse needs of learners. Modern educational environments need intelligent methods of continuous assessment of learners' achievements and personalized instructional pathways. Therefore, modern studies have focused on the application of Artificial Intelligence for student assessment and learning.

Artificial Intelligence in Student Assessment

Artificial Intelligence (AI) has brought a complete transformation in educational assessment due to its capability for continuous evaluation and personalized analysis of student learning. Unlike conventional methods of assessing student performance, which are limited to measuring performance at specific times, the use of an AI-powered assessment system continuously gathers and analyzes students' data throughout the learning process. Continuous assessment allows the teachers to pinpoint and understand the needs and strengths of learners. As educational technology advances, AI has been incorporated into modern education assessment to support adaptive learning.

The practical application of AI in students' assessment involves the utilization of intelligent evaluation algorithms to evaluate the student outcomes and recommend a proper educational intervention. For instance, Shukla et al. (2021) suggest that AI-assessment frameworks evaluate the students' response, establish patterns of comprehension, and propose the necessary instructional intervention depending on the learners' mastery of knowledge. Unlike the typical approach to grading systems, this computing model allows not only personalization but also formative assessment. Additionally, the authors integrate Markov Decision Processes (MDPs) into their framework, which helps to choose the optimal learning activities for the students according to their current knowledge status.

In addition to AI advancements, there is another technology that allows estimating learner competency and mastery through their progress in learning tasks, called knowledge tracing techniques. According to Šarić-Grgić et al. (2024), Bayesian Knowledge Tracing (BKT) is a technique that estimates learner mastery based on learners' previous test scores and correct or wrong answers. Likewise, Zhou et al. (2025) concluded that Deep Learning Knowledge Tracing (DLKT) is a method that helps predict learners' progress more accurately through deep learning. Knowledge tracing has become an important element of AI-based adaptive learning environments.

Despite the many strengths AI provides in the field of educational assessments, there are several challenges associated with the implementation of such an assessment. According to Shukla et al. (2021), some of the challenges

include issues related to computation, transparency of algorithms, privacy of data, and availability of enough learner data. It is important to address these challenges so that the AI-driven assessment system continues to be reliable and accurate. Despite these weaknesses, further developments in technology have helped enhance the accuracy and efficiency of AI-driven assessment systems.

With the constant advancements being made to AI assessment technologies, there is now a need for educational systems to have intelligent assessment content for use in continuously assessing learners. Adaptive learning systems need many assessment items to be able to assess the progression of the learners in their learning process. The need for such assessment content has prompted researchers to focus on finding ways to generate such assessments through artificial intelligence. Automatic generation of assessment questions eases the load of teachers and enables continuous assessment.

Automatic Question Generation Using Artificial Intelligence

The increasing usage of digital learning environments has resulted in an increase in the need for an efficient method of assessment. Traditionally, it takes a considerable amount of time and effort to create assessments, and the educator is supposed to have sufficient knowledge about the topic. Therefore, researchers have tried to find means through which they can use artificial intelligence (AI) to automatically generate questions.

As per Kurdi et al. (2020), Automatic Question Generation (AQG) means the process of creating assessment questions by employing computational methods from structured or unstructured educational material. This approach to question creation was introduced to cater to the challenges faced in developing questions manually and in performing different educational tasks like formative assessments, self-assessment, adaptive learning, and online teaching. The importance of AI-enabled question generation is that it helps educational organizations to consistently generate assessment questions and minimize the effort of teachers.

Apart from increasing efficiency, another benefit that AQG brings along is the improvement of the learning process through reinforcement and assessment. As stated by Kurdi et al. (2020), automatic generation of questions can be used in the retrieval of information, improvement of the concept, detection of misconceptions, and constant assessment in the learning process. Due to the increased application of e-learning systems, AQG has become increasingly important because of its ability to form large assessment databases.

In today's Automatic Question Generation (AQG) systems, the Natural Language Processing (NLP), machine learning, and semantics techniques are employed to convert educational material into valuable test questions. The use of these methods will help the AI to understand educational content, recognize relevant ideas and concepts, and produce tests in various formats, including multiple-choice questions, fill-in-the-blank questions, and short answers. As

instructional materials are mostly available in text format, text-based AQG continues to be one of the most applicable methods of AQG.

These theories, provided by Kurdi et al. (2020), form a solid theoretical background for the application of Automatic Question Generation (AQG) in the educational environment. In the proposed LearnQuest system, it is intended to implement automatic generation of questions based on artificial intelligence that aligns with the educational content by the teacher. Using quiz automation, the system minimizes the burden of work on teachers while delivering the exams to the students on time. The responses obtained in the AI-generated quiz are used for Bayesian Knowledge Tracing, allowing the system to determine how well the student knows the subject matter. This achievement is the reason for making AQG one of the most important AI functions used in LearnQuest.

While there are many benefits associated with AI-based assessment tools, the education sector must ensure that these results are easily understood by both learners and educators. Visualization plays a key role in helping learners and educators make sense of learning analytics promptly. Consequently, recent studies have explored the use of radar charts as an effective tool for presenting multidimensional learner performance.

Radar Chart for Skill Visualization

Radar Charts are widely used in the educational field because they serve as a means of illustrating how students perform in different academic performances in a proper and easy to read and understand chart. Several studies by Jayaprakash et al. (2024), Kantabutra et al. (2025), Ramaswami et al. (2023), and Pinargote et al. (2024) illustrate that properly designed radar charts assist in detecting the weaknesses and strengths of students and learning tendencies.

This study employs radar charts in the LearnQuest system to show students' degree of mastery of competencies in science subjects. The assessment results will provide students with a perspective on how they perform because of the main science competencies and allow students to assess their own level of knowledge while helping teachers make evaluations of students' performance quickly.

Visualization methods have been quite successful in communicating learner performance, but there needs to be an intelligent system that can interpret learner performance information and make personalized recommendations based on their performance. All these requirements are achieved by embedding Artificial Intelligence services into Learning Management Systems using Application Programming Interfaces (APIs).

AI API Integration for Adaptive Learning and Intelligent LMS

The implementation of Artificial Intelligence (AI) Applications Programming Interface (APIs) has greatly improved the functionality of the current Learning Management Systems (LMS) by providing intelligent education capabilities without the need for institutions to create AI models from the beginning. The use of AI APIs allows educational systems to integrate into their operations such elements of advanced Artificial Intelligence as auto-generation of content, personalized feedback, assessment, and intelligent support of learners. According to Zawacki-Richter et al. (2020), the application of AI technologies into educational platforms make possible to automate various instruction-related procedures and supports provide personalized learning experiences due to the analysis of students' activities and learning materials. In the same way, Holmes et al. (2022) pointed out that AI-enabled systems can offer an adaptive learning experience, make better instructional decisions, and facilitate learning support for students through their unique requirements.

Additionally, the advancements made in Large Language Model (LLM) have also enhanced the abilities of the AI-powered educational system. According to Kasneci et al. (2023), the integration of LLMs through APIs into Learning Management Systems enables the provision of timely, scalable, and intelligent educational services, which involve content creation, automated evaluation, and personalized feedback that adapts to the performance of learners.

The literature above clearly shows that the use of artificial Intelligence (AI) APIs is a firm basis for the establishment of an intelligent adaptive learning management system (LMS). In the process of combining AI and learning analytics, one can automate student support, improve the generation of quality content, and ensure the customization of any learning process. This directly aligns with the goals of LearnQuest, an adaptive learning management system powered by Artificial Intelligence.

To achieve this, LearnQuest utilized a Llama 3.3-70B-Instruct model via an API to automatically provide lesson summaries, evaluation questions, and custom learning paths based on recent uploads. On the other hand, to justify the outputs, the AI system uses Bayesian Knowledge Tracing (BKT), which enables it to manage each student's mastery and initiate the relevant adaptive actions. In the end, it has provided LearnQuest with a data-oriented service.

Synthesis

It clearly demonstrates in the review literature how educational technology has evolved extremely in the implementation of the Learning Management System (LMS), Artificial Intelligence (AI), Automatic Question Generation (AQG), learning analytics, Radar Charts, and AI Application Programming Interfaces (APIs). It can be concluded that all these improvements have changed the traditional digital learning platforms into an intelligent environment that can support the process of

continuous assessment and evaluation of learners' progress. In the modern educational system is not only about imparting knowledge; rather, it is a platform that can help to increase efficiency in teaching and improve the learning outcomes of the students.

The study of learning management systems (LMS) provides the technological foundation for digital learning since it highlights the significance of engaging the learner, ensuring accessibility, and using learning analytics to increase the quality of instruction (Bonilla-Priego, 2024; İskigöz, 2024). These abilities enable integration of AI to help achieve continuous assessment, competency tracking, and personalized feedback. At the same time, research conducted concerning AI-assisted assessment of students emphasizes the ability of intelligent systems to evaluate student performance, determine their level of competency mastery, and deliver instruction at the right time (Shukla et al., 2021; Šarić-Grgić et al., 2024; Zhou et al., 2025).

Additionally, the discussion on Automatic Question Generation (AQG) reveals the ways of employing artificial intelligence (AI) for the automation of assessment creation with the Natural Language Processing (NLP) methods, and machine learning can provide teachers with a way to lessen their difficulties and workload in conducting the assessment process (Kurdi et al., 2020). The effectiveness of these intelligent assessments is enhanced by employing visualization tools like radar charts to help teachers and students better understand the data about different types of competences (Jayaprakash et al.; Kantabutra &

Tangmanee, 2025; Ramaswami et al., 2023; Pinargote et al., 2024). In addition, the use of AI through application programming interfaces (APIs) allows for the creation of platforms to provide smart services like lesson summaries along with customized feedback and suggestions, but ensuring that the system architecture is scalable and efficiency-compliant (Gupta et al., 2022; Holmes et al., 2022; Santos et al., 2021).

In summary, the reviewed literature supports both theoretically and technologically proposed LearnQuest: An Adaptive Learning Management System Powered by Artificial Intelligence for Grade 11 and Grade 12 STEM Students of Kapitolyo High School. The findings indicate that the use of an LMS combined with artificial intelligence assessment, automatic question creation, learning analytics, radar chart representation, and integration of AI APIs could help considerably enhance the process of competency monitoring and personalize the learning process. These ideas will form the basis for further discussion of adaptive learning algorithms such as Bayesian Knowledge Tracing (BKT), Deep Learning Knowledge Tracing (DLKT), and Markov Decision Processes (MDP).

Personalized Learning Using Artificial Intelligence

The application of AI in education has led to the development of personalized learning, considered one of its most important uses in the transformative application of AI in modern educational fields. Personalized

learning enables the use of instructional methods that suit the learning requirements and capabilities of students based on their unique abilities, learning needs, and performance progress of the students. In contrast with traditional classroom instruction, where every student learns through uniform study materials and assessments, regardless of their knowledge levels, AI-based personalized learning examines students' data on an ongoing basis and offers personalized recommendations on materials, assessments, and educational routes that facilitate the process of learning.

As pointed out by Khosravi et al. (2022), AI-based personalized learning systems use the information obtained from learner interactions, assessments, and behavior analysis to design personalized learning experiences for individuals. The intelligent systems under consideration can analyze students' academic performance, identify areas of weakness, and propose relevant learning sources appropriate for the students' level of knowledge. Hence, learners can learn at their own speed while teachers receive essential information that ensures effective support of students.

Likewise, Zawacki-Richter et al. (2020) elaborated that Artificial Intelligence improves personalized learning by offering adaptive learning materials, automatic feedback, learning suggestions, and learning analytics that meet the specific needs of learners. Such AI-based abilities solve the problems associated with the traditional one-size-fits-all approach to instructions by offering education that is adaptive, flexible, and personalized. By continuously analyzing

the performance of learners, Artificial Intelligence can help educational systems in modifying instructional assistance as well as recommending suitable learning resources for the continuous improvement of outcomes.

In the recent educational studies (Khosravi et al., 2022; Zawacki-Richter et al., 2020), it must be highlighted that personalized learning outcomes are crucial for the development of self-regulated and independent learners. When educational content and activities are delivered in accordance with the students' level of competence, identify the strong and weak areas of the students and actively monitor their academic performance. Therefore, apart from enhancing learners' academic performances, personalized learning helps to develop their autonomy, motivation, and lifelong learning skills.

A thorough analysis of the literature shows that Artificial Intelligence significantly contributes to the creation of personalized learning environments that adjust their teaching methods in accordance with actual student performance and progress in gaining skills (Khosravi et al., 2022; Zawacki-Richter et al., 2020). These findings strongly support the proposed LearnQuest: An Adaptive Learning Management System Powered by Artificial Intelligence, which uses AI to provide personalized learning experiences through summaries of lessons, AI-based assessments, personalized feedback, and competency-based learning suggestions. To make learning more adaptive, Bayesian Knowledge Tracing (BKT) is used in the system to predict the mastery level of the learners based on their performance in the assessments. The LearnQuest system will then

recommend suitable learning materials and interventions to the students to meet their learning requirements. In this way, making the process more adaptive and learner-centric for Grade 11 and 12 STEM students of Kapitolyo High School.

AI-Based Lesson Summarization Using Artificial Intelligence

Utilization of Artificial Intelligence (AI) in the application of Large Language Model (LLM) has clearly reshaped the way of delivering and consuming in the online educational environment. AI is one of the crucial applications that can summarize the lessons that can automatically compress the length of educational materials into meaningful and easy to understand, while maintaining the original concept. As Zawacki-Richter et al. (2024) note, the artificial intelligence-driven technologies facilitate the development of intelligent content that improves the learning efficiency, accessibility, and personalization of students.

The new development of LLMs has enabled AI-generated content to improve the quality of summarization of lessons through understanding the context and creating meaningful content, rather than pulling out the important information. Xie et al. (2025) stated that AI-assisted lesson summarization provides students with the ability to concentrate on important aspects of lesson materials, thereby improving comprehension, the ability to recall the knowledge, and learning satisfaction. And with this outcome, AI summarization can support the progression of teaching and self-study.

Additionally, AI-based lesson summaries can help in personalized learning through the provision of educational content in an organized manner. According to Xie et al. (2025) and Zawacki-Richter et al. (2024), rather than being a replacement for teachers, AI is an important tool that can support the learning and students in reviewing their lessons in a better way while teachers concentrate on facilitating their learning.

According to the literature review (Xie et al., 2025), learning analytics plays an important role in monitoring the performance of students as well as helping in their education process. Considering these results, the proposed system is LearnQuest: An Adaptive Learning Management System powered by Artificial Intelligence, where the Llama 3.3-70B-Instruct model was implemented using an API for creating lesson summaries based on uploaded educational content. The summaries generated by AI are useful for Grade 11 and 12 Kapitolyo High School STEM students, allowing them to quickly review necessary concepts before assessments, thus ensuring that learners understand the study material and adding to the efficiency of the system.

Learning Analytics for Monitoring Student Performance

Learning Analytics has become basis elements of an intelligent educational system through empowering teachers to collect, analyze, and interpret the data of their students to improve their teaching effectiveness and student achievement. In

their comprehensive review, Viberg et al. (2020) observed that the current Learning Management System (LMS) keeps tracking and recording the students' interactions, assessments, and participation outcomes, and the student behavior in the learning process, which can be easily used to make an educational decision based on the evidence and to identify the students who need specific intervention.

Focusing on the dashboard implementation, Matcha et al. (2022) found that the method of having good monitoring techniques using an analytics dashboard whereby students can be tracked in terms of their performance, engagement, and skill development. Technology helps teachers in identifying the learning gaps, determining the effectiveness of the learning process, and addressing individual students' learning needs.

Looking at the advanced system, Ifenthaler et al. (2023) noted that learning analytics form the fundamental basis of adaptive learning technologies. Analytics systems provide the data that Artificial Intelligence needs to prepare personalized feedback, which checks proficiency and facilitates decisions regarding instruction analytics technology. Through combining the integration of learning analytics and artificial intelligence, it becomes possible for the educational system to move beyond performance measurement and provide intelligent, automated, and adaptive learning.

In conclusion, the combined perspectives of Viberg et al. (2023), Matcha et al. (2022), and Ifenthaler et al. (2023) indicate that learning analytics is crucial for tracking student progress and facilitating personalized education. These outcomes

strongly support the proposed system, LearnQuest: An Adaptive Learning Management System Powered by Artificial Intelligence that harnesses learning analytics through AI to evaluate the learning results and assess the established competencies. LearnQuest provides these insights into performance through interactive dashboards that enable the students in Grades 11 and 12 of Kapitolyo High School to monitor their learning styles independently. Besides, these analytics also play a crucial role in the entire AI lesson summarization process and automated question generation performed by the system; as a result, every learning intervention is based on data and adjusted to the individual needs of students.

Competency-Based Learning and Mastery Learning

Competency-Based Learning (CBL) is a type of learning approach that focuses on students showcasing certain competencies in terms of knowledge and skills, with the students becoming the focus instead of time spent on learning. Under this approach, students are not allowed to move forward until the student mastered the necessary skills through continuous assessment and educational assistance. As per UNESCO (2021), competency-based learning includes defined learning outcomes, continuous tracking of students' progress, and approaches to instruction that ensure that students can demonstrate the required skills before moving to advanced topics. In the same vein, Sturgis et al. (2021) maintain that

mastery learning enables students to progress according to their own pace, along with timely assessments and support.

The use of competency-based learning has been shown by UNESCO (2021) and Sturgis et al. (2021) to be advantageous to the behavior of students. Through granting students concentrate entirely on gaining competencies instead of rushing through rigid course schedules, there will be increased ownership of their learning process by the students. Continuous assessment and competence monitoring offer teachers valuable, detailed information that helps in making active decisions about instruction and allows personalized intervention in real-time depending on the actual needs of learners.

In conclusion, the theories provided by UNESCO (2021) and Sturgis et al. (2021) make an excellent basis for the proposed system, LearnQuest, an adaptive learning management system powered by artificial intelligence. The system uses the basic principles of competency-based learning in that it incorporates the Llama 3.3-70B-Instruct model using the AI API to produce lesson summaries, automatic evaluation and feedback, and competency-based learning suggestions based on the uploaded educational content. Moreover, LearnQuest makes use of the Bayesian Knowledge Tracing (BKT) algorithm to dynamically assess the students' competence in mastering the skills through these assessment tools. The intelligent features allow LearnQuest to precisely identify the gaps in learning, measure progress, and suggest appropriate learning interventions for achieving mastery learning among the STEM Grade 11 and 12 learners of Kapitolyo High School.

With an adaptive educational system using AI to personalize learning, it is important to select suitable algorithms that can estimate the mastery of competency. Hence, the succeeding section presents a comparative analysis of Bayesian Knowledge Tracing (BKT), Deep Learning Knowledge Tracing (DLKT), and Markov Decision Processes (MDP) to identify the most suitable algorithm for the adaptive learning model in LearnQuest.

Comparative Analysis of Adaptive Learning Algorithm of BKT, DLKT, and MDP

The adaptive learning management system depends on Artificial Intelligence (AI) algorithms for monitoring the progress of students, assessing the mastery of competencies, forecasting academic performance, and facilitating personalized instruction. The performance effectiveness of such systems depends on choosing a suitable learner modeling algorithm that effectively captures the process of learning by students. Some of the adaptive learning algorithms that have been extensively used include the Bayesian Knowledge Tracing (BKT), Deep Learning Knowledge Tracing (DLKT), and the Markov Decision Process (MDP), which could provide different strengths and limitations.

Bayesian Knowledge Tracing (BKT) assesses competence mastery of students through the pattern of correct and incorrect answers. As stated by Šarić-Grgić et al. (2024), BKT is quite efficient in working with limited student interaction

of data and provides understandable and transparent probability values that can be interpreted easily by teachers. In contrast, according to Zhou et al. (2025), the Deep Learning Knowledge Tracing (DLKT) applies deep neural networks in order to detect intricate learning patterns and make accurate predictions about students' performance in the future, provided enough educational data is available. However, it also highlighted that DLKT requires much more computing power and is less interpretable than BKT.

While BKT and DLKT aim at measuring the students' proficiency, MDP concentrates more on selecting the best instructional actions than estimating students' proficiency. Zhou et al. (2022) noted that MDP represents the learning process using states, actions, and rewards to suggest the best instructional intervention. Though useful in adaptive instructional planning, it requires enough student interaction data for successful applications, hence making it less appropriate during the initial phase of the system deployment.

Overall, the collective findings from Šarić-Grgić et al. (2024), Zhou et al. (2025), and Zhou et al. (2022) show that each of the algorithms for adaptive learning has specific advantages and is based on the objectives of the educational system. The BKT model is ideal for use in monitoring the competence of students due to ease of use, minimal data needs, and easy interpretability. While MDP is more suitable for long-term instruction decision-making and adaptive processes, these major differences can be seen in Table 1.

Table 1*Comparison of Adaptive Learning Algorithms*

Features/ Algorithm	Bayesian Knowledge Tracing (BKT)	Deep Learning Knowledge Tracing (DKLT)	Markov Decision Process (MDP)
Primary Focus	Estimate competency mastery levels	Predict future performance	Selecting Optimal instructional actions
Data Requirement	Low -Suitable for early system use	High - Large- scale data	Moderate to High- Require extensive interaction logs
Interpretability	High	Low	Moderate
Best for	Real-time classroom adaptivity	Long-term forecasting	Adaptive instructional decision-making

According to the theories mentioned above, most literature supports the view that the Bayesian Knowledge Tracing (BKT) algorithm is best suited for LearnQuest, An Adaptive Learning Management System Powered by Artificial Intelligence. The uniqueness of BKT is to offer students precise, interpretable, and real-time measures of their competency, making it useful for the purposes of

mastery tracking, personalized educational feedback, and adaptive learning for grade 11 and 12 STEM students in Kapitolyo High School.

Synthesis

The reviewed literature reveals that AI makes an important contribution to creating learner-oriented education systems through personalized learning, AI-based lesson summarization, learning analytics, competency-based learning, and adaptive learning algorithms. Collectively, these studies show that today's education technologies extend further than delivering educational content, as they provide an intelligent tool that allows for continuous assessment of students' performance and develops appropriate assistance for learning.

In the literature regarding personalized learning, it is emphasized that artificial intelligence can monitor the learner and evaluate learning behavior to offer a customized learning experience, as well as give useful feedback on competency (Khosravi et al., 2022; Zawacki-Richter et al., 2020). Likewise, the studies on lesson summarization powered by AI technology demonstrate that large language models allow the generation of effective summaries from the learning materials, which contribute to better comprehension and memorization of information, as well as self-studying (Xie et al., 2025; Zawacki-Richter et al., 2024).

The research mentioned confirms the significance of learning analytics in determining student performance, as it helps in transforming learner behavioral

data into useful information that can be utilized in decision making, competence evaluation, and personalizing learning (Ifenthaler & Schumacher, 2023; Matcha et al., 2022; Viberg et al., 2020) Learning analytics is made more significant by competency-based learning, which deals with an continuous assessment, reaching learning goals, and personalizing learning before moving learners to advanced competencies (Sturgis & Patrick, 2021; UNESCO, 2021). Moreover, the comparison between Bayesian Knowledge Tracing (BKT), Deep Learning Knowledge Tracing (DLKT), and Markov Decision Process (MDP) shows that each algorithm of adaptive learning has its own benefits according to the purpose of the system, available information about learners, and the implementation needs. Out of all the methods mentioned above, Bayesian Knowledge Tracing (BKT) is the most appropriate method due to the combination of all the mentioned factors (Šarić-Grgić et al., 2024; Zhou et al., 2025; Zhou & Wang, 2022).

In conclusion, the literature presented above offers a detailed theoretical and technological basis for the system being proposed, LearnQuest: An Adaptive Learning Management System Powered by Artificial Intelligence for Grade 11 and Grade 12 STEM Students of Kapitolyo High School, and the results highlight the importance of AI-based personalized learning, lesson summaries, learning analytics, competency-based-learning, and Bayesian Knowledge Tracing (BKT) in the context of monitoring student performance, providing recommendations, and improving their experiences. In this respect, all these concepts serve as the theoretical foundation for creating an intelligent and adaptive Learning

Management System that supports competency-based education and enhances STEM students' academic performance.

Significance of the Study

The implementation of LearnQuest, An Adaptive Learning Management System Powered by Artificial Intelligence for STEM Students of Kapitolyo High School, is expected to be advantageous to the following parties:

Kapitolyo High School. The deployment of the LearnQuest will enable it to more efficiently organize, deliver, and manage the learning resources for its STEM strand. The use of artificial intelligence by schools can increase students' engagement levels, improve learning, and help students achieve academic success. In addition, data-driven analytics will help the administrator at the school to gather useful insights concerning the performance of the curriculum and students.

Grade 11 and 12 students. These students stand to gain much from having an adaptive learning environment customized for them. Using the AI features in LearnQuest, students will be able to monitor their learning progress, receive personalized lesson summaries, and undergo automated testing based on their competencies. Through personalized feedback, the students will be able to recognize their strengths and weaknesses, thereby adopting good study habits.

The Teachers and Educators. The proposed system will help in mitigating the existing administrative and educational problems that teachers encounter using AI-based learning analytics. Through real-time tracking and performance reporting, the system will enable teachers to identify learners who are having difficulties in their studies and offer academic intervention. The question-generating feature will simplify the evaluation process for the teachers and allow them to make necessary changes to the instructional process.

The Researchers. Conducting this research gives student researchers an opportunity to use the theoretical knowledge of software engineering, database management, artificial intelligence, and user experience (UX) in solving a real-world problem. This presentation provides them with practical skills of technology, systems integration, and project management that are relevant for their future work in the sphere of information technologies.

Future Researcher. This study is meant to act as an empirical guide for researchers seeking knowledge about the subjects of adaptive learning systems and educational technology. These research methods and findings can be employed to investigate even more advanced uses of artificial intelligence, apply the same system in other subjects, or perform longitudinal research regarding the influence of artificial intelligence on students' results.

Statement of the Problem

For this study to develop LearnQuest, an adaptive learning management system powered by artificial intelligence for STEM students of Kapitolyo High School, it is important to identify the issues that will be addressed by the system.

General Problem

How can an artificial intelligence-powered adaptive learning management system improve the learning outcomes, competency mastery levels, and academic performance of Grade 11 and Grade 12 STEM students in science subjects at Kapitolyo High School?

Specific Problems

1. What system features and functionalities should be developed for the LearnQuest to help students in science subjects in material handling, processing of materials using AI technologies, quizzes and assessments, progress tracking, and personalization of learning experiences?
2. How can a Large Language Model (LLM), specifically the Llama 3.3-70B-Instruct model, be integrated into the LearnQuest system to generate science-related quiz questions, summarize instructional materials from PDF documents and PowerPoint presentations, and provide personalized feedback to students?

3. How can Bayesian Knowledge Tracing (BKT) and learning analytics be applied in the LearnQuest system for estimating students' level of mastery, recognizing learning gaps, and facilitating adaptive learning methods in science competencies?
4. What learning analytics and dashboard features can be implemented in the LearnQuest system to monitor student performance, visualize mastery level, identify at-risk students, and assist the teacher in decision-making?
5. How acceptable is the developed LearnQuest system as evaluated by students and teachers based on the ISO/IEC 25010 Software Quality Model in terms of:
 - a. Functional Suitability;
 - b. Performance Efficiency;
 - c. Compatibility;
 - d. Usability;
 - e. Reliability;
 - f. Security;
 - g. Maintainability; and
 - h. Portability?

Scope and Limitations of the Study

For this study to develop LearnQuest, an adaptive learning management system powered by artificial intelligence for STEM students of Kapitolyo High School, it is important to identify the issues that will be addressed by the system.

Scope of the Study

Learning Management Features

The LearnQuest system provides a web-based learning platform that allows teachers to upload, organize, and manage science learning materials, while enabling students to access instructional content in a centralized and structured environment. This supports improved accessibility and organization of learning resources.

Supported Learning Materials

The system is designed to process instructional content resources in PDF documents and Microsoft PowerPoint presentations. These documents will be used as the basis for AI-enabled production of learning content.

Artificial Intelligence Integration

The LearnQuest system implements the Llama 3.3-70B-Instruct Large Language Model (LLM) by using the inference API services for adaptive learning

features. The mode analyzes the uploaded instructional content and student performance records to:

- Automatically analyze uploaded learning materials.
- Generate lesson summaries.
- Generate interactive quizzes and assessment questions.
- Provide personalized feedback to students.
- Automatically generate learning recommendations.
- Support adaptive learning experiences based on the estimated level of students using Bayesian Knowledge Tracing (BKT)

Assessment and Adaptive Learning Features

The LearnQuest system incorporates adaptive mechanism features includes:

- AI-generated quizzes for specific science competencies.
- Competency-based evaluation activities.
- Adaptive assessment generation based on student mastery levels.
- Personalized learning pathways based on learning performance.
- The teachers provide the learning material, and the system generates the assessments automatically from the uploaded material.

Mastery Tracking and Learning Analytics

The LearnQuest system applies Bayesian Knowledge Tracing (BKT) to estimate the mastery level of students on science competencies, identify the learning gaps and deficiencies, and track the progress of the student over time in relation to the study objective of improving the competency mastery. The BKT model utilizes four parameters:

- **P(L₀)- Initial Probability of Mastery:** The probability of knowing the competency before answering the first question.
- **P(T)- Probability of Learning:** The probability of moving into a state of mastery after trying to answer a question.
- **P(G)- Probability of Guessing:** The probability of answering correctly without mastering competency.
- **P(S)- Probability of Slipping:** The probability of answering wrongly even though the student is in a state of mastery.

Student Learning Data Collected

The system LearnQuest will collect the following learner data in accordance with analytics and BKT needs of the system:

- Quiz responses
- Question-level correctness
- Assessment history
- Competency mastery records

- Learning activity completion records
- Generated quiz performance
- Student learning progress

Monitoring and Reporting Features

To assess learning outcomes, competency attainment, and performance of the students, the system offers dashboards and analytics, including:

- Student progress dashboards
- Competency mastery dashboards
- Performance analytics
- Learning trend reports
- Identification of students requiring intervention
- Personalized study recommendations
- Adaptive learning pathway recommendations

System Evaluation

The developed system shall be evaluated by selected Grade 11 and Grade 12 STEM students and Science teachers of Kapitolyo High School using the ISO/IEC 25010 Software Quality Model in terms of:

- Functional Suitability
- Performance Efficiency
- Usability

- Reliability
- Security
- Maintainability
- Portability
- Compatibility

Limitations of the Study

This study is subject to the following limitations, in line with its objectives and scope:

Academic Scope

The study is limited to science subjects for Grade 11 and Grade 12 STEM students of Kapitolyo High School, and the results gathered from the respondents cannot be generalized institution-wide without further validation.

Artificial Intelligence Limitations

The Artificial Intelligence component of LearnQuest is limited to the capabilities of the Llama 3.3-70B-Instruct Large Language Model. While the model can generate educational content and support adaptive learning, its performance is subject to the following limitations:

- The AI generates lesson summaries, quizzes, personalized feedback, and learning recommendations based solely on the

content of the uploaded instructional materials. The quality and accuracy of the generated outputs depend on the completeness, clarity, and relevance of the uploaded learning resources.

- AI-generated summaries may omit important concepts, oversimplify complex scientific topics, or produce explanations that require further clarification by the teacher.
- AI-generated quiz questions may occasionally contain ambiguous wording, factual inaccuracies, duplicate questions, or questions that do not fully align with the intended learning competency. Therefore, generated assessments may require teacher review and validation before classroom use.
- Personalized feedback and learning recommendations are generated using the student's estimated mastery levels and recorded learning data. These recommendations are intended to supplement instruction and may not fully account for individual learning preferences, classroom participation, laboratory performance, or other non-system factors that influence learning.
- The AI model does not independently determine student grades, competency mastery, or academic standing. Mastery estimation is performed using Bayesian Knowledge Tracing (BKT), while the AI utilizes the resulting mastery information to generate adaptive content and recommendations.

- The AI model relies on the information provided within the uploaded learning materials and does not independently verify scientific accuracy or consult external educational references during content generation. Consequently, teacher supervision remains necessary to ensure that generated instructional materials are pedagogically appropriate and factually correct.
- The system functions solely as an instructional support tool intended to enhance teaching and learning. It is not designed to replace teacher expertise, professional judgment, curriculum planning, or assessment validation.

Mastery Estimation Limitation

The study has limitation estimation on mastery level of students includes:

- Mastery estimation is based on Bayesian Knowledge Tracing (BKT) and learner interaction data.
- The accuracy of mastery estimates depends on the quantity and quality of collected assessment data.
- Competency mastery levels are estimates and may improve as more student interaction data become available.

System Scope Limitation

The LearnQuest system does not include:

- Fully autonomous tutoring systems
- Speech-based learning interfaces
- Virtual reality or augmented reality learning environments
- Mobile application development
- Integration with External Learning Management Systems
- Advanced predictive analytics models beyond BKT-based mastery estimation
- Automated grading of essay-type or subjective assessments

Generalizability Limitation

The outcome of this research is confined to the setting where Kapitolyo High School is implemented. Consequently, the results cannot be applied directly to other schools without additional studies.

CHAPTER 2

METHODOLOGY

Introduction

This chapter provides information on the research methods and processes used in developing LearnQuest: The Adaptive Learning Management System Powered by AI for STEM Students of Kapitolyo High School. It provides the processes that were followed in the research, including the design, respondents, methods of collecting data, development methodology of the system, instruments used for evaluation of the system, and methods of analysis.

The methodologies used in this study have been chosen to address the issues mentioned in Chapter 1, such as the absence of adaptive feedback, personalized assessment, and learning analytics from existing Learning Management Systems (LMS). The use of these methodologies ensures an organized process of creation, implementation, development, and evaluation of the proposed artificial intelligence-based adaptive learning management system.

In addition to this, this chapter provides information about the way that the developed system would be evaluated because of its functionality, usability, software quality, and acceptance. By systematically describing the research procedure, this chapter ensures that the research follows well-established principles of developmental research.

Research Design

The study involved the use of a development research design that relied on descriptive research approaches. The main aim of this research is to create, develop, implement, and assess LearnQuest: An Adaptive Learning Management System Powered by Artificial Intelligence for STEM Students of Kapitolyo High School. The suggested model aims at providing personalized learning, adaptive assessment, learning analytics, and competency-based feedback for Grade 11 and Grade 12 STEM students.

The development component of the research is about designing, developing, implementing, and improving the system known as LearnQuest using iterative planning, testing, evaluation, and improvement. The system development has followed the Agile Software Development Methodology, which enables continuous improvement of features of the system based on evaluation outcomes and user feedback.

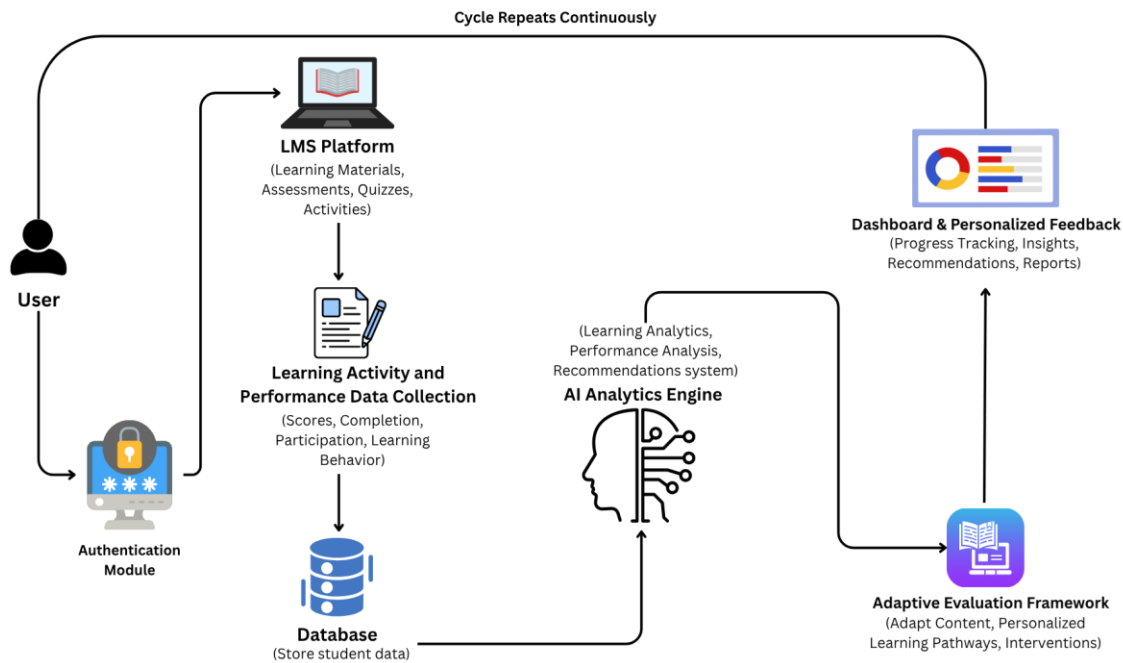
The AI-generated components of the system utilize the Llama 3.3-70B-Instruct Large Language Model (LLM) to create summaries of lessons, quizzes, feedback, and suggestions for the learner from the provided instruction material. The AI-generated information is used together with learning analytics and BKT mastery calculations to provide adaptive learning and competency-based suggestions for the learner.

This system will be evaluated based on the ISO/IEC 25010 Software Quality Model with respect to Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Additionally, the acceptability of the proposed system will be measured by respondents. The outcomes of the evaluation will be utilized to measure the effectiveness, software quality, and usability of LearnQuest as an Artificial Intelligence-driven adaptive learning management system.

As stated by Aljad et al. (2025), the development of the modern Learning Management System necessitates a user-centered and iterative process that allows for continuous improvements of the system, evaluations of its performance, and implementation of advanced technologies, including Artificial Intelligence. The use of the specified process is well justified in the current research because it ensures the development of the learning management system powered by artificial intelligence that can evaluate its potential to improve the learning experience with adaptive assessments, feedback, and learning analytics. Moreover, the use of Artificial Intelligence will enhance the adaptability of the system to meet the needs of STEM students.

Figure 1

Architectural Framework of the Study



Architectural Framework

The architectural framework is designed to propose a workflow of LearnQuest, an adaptive learning management system powered by artificial intelligence for the STEM students of Kapitolyo High School. The proposed Learning Management System is a personalized and data-driven adaptive Learning Environment that will help meet the academic needs of Grade 11 and Grade 12 STEM students. This will enable effective interaction among students, teachers, and learning resources using AI-powered analytics to enhance learning outcomes; provide real-time monitoring of student performances; and achieve personalized support for each student within a safe and secure Learning Environment.

The students will use the system when they sign in to initiate the learning process. The system has an Authentication Module in place that verifies the user's credentials to allow access to the Learning Management System (LMS). Through the Authentication Module, the user is securely authenticated for communicating secure data through secure transmission, and the users have been validated before gaining access to the LMS, which will protect the student and the data from being improperly used throughout their learning experience.

Upon successful authentication, users are transferred to the LMS Platform, which is essentially the core part of the entire system. The learning management system (LMS) allows students to browse through digital learning materials, assessments, quizzes, assignments, and instructional resources related to their STEM subjects. Teachers can upload lesson plans, observe their students' actions, and manage the content for all their courses from the LMS. The LMS is the system's main source for all student learning activities and serves as a link between the student learning activities and the system's data collection and AI components.

The Learning Activity and Performance Data Collection Module records all learning activities recorded as students engage in the platform. This module collects academic data such as quiz scores, assessment scores, completion rates for each lesson, participation records, and patterns of learning. The data collected provides insight into how well students are learning and is transmitted into the system's database continuously for storage and further analysis.

All data associated with learning will be protected by the database, which serves as the repository of learning-related data. The database contains profiles of students, records of assessments, logs of learning activities, histories of academic performance, and AI-generated suggestions related to learning from assessments or other student information. Furthermore, the database ensures the integrity of all data kept within it, retrieves or accesses its data efficiently from any part of the entire system, and keeps all parts of the system synchronized in order to provide uninterrupted services.

An AI analytics engine, which uses artificial intelligence techniques (such as learning analytics and performance analysis) to help identify patterns of student learning behavior, track academic progress, and discover problems with students' skills to make recommendations. The AI analytics engine will evaluate student performance data to provide an understanding of performance characteristics, areas of strength, weakness, and opportunities for improvement. By analyzing student data, personalized recommendations for support will be provided to meet each student's individual needs through adaptive learning technologies.

An AI Analytics Engine's Key Feature involves using Bayesian Knowledge Tracing (BKT) to assess students' levels of mastery in identified areas of learning. To accomplish this, the system gathers assessment data and previous learning activity from individual students to compute the probability that the student has attained mastery for each skill and concept. This will also allow for the ongoing monitoring of students as they continue their rate of learning, determining when to

provide necessary assistance according to the level at which the learner has reached.

The Adaptive Evaluation Framework supports the Intelligent Analysis Process of the AI Analytics Engine through output from the engine to provide a set of adaptive learning recommendations, or personalized learning pathways and intervention strategies based on a learner's performance and the academic needs of each learner. Instead of relying only on static assessments, the Adaptive Evaluation Framework will provide continuous evaluation of the learner's progress and adaptive decision-making support for continuous improvement of the learner's STEM conceptual mastery. The output produced by the AI components is displayed through both the Dashboard and the Personalized Feedback Module. The dashboard provides the student and the teacher with a current visual representation of data, such as Learning Progression, Performance Trend, Mastery Level, and Learning Action Recommendations. Students can receive personal feedback, study suggestions, and progress reports to help them achieve their educational ambitions. The educator has access to a dashboard that allows tracking of student performance, identification of students who may need help, and making evidence-based instructional decisions.

The whole system runs continuously through each of its phases; this allows it to obtain new data for learning, update its analysis by Artificial Intelligence (AI), generate recommendations adapted to the learning style of the student, and

develop the responsiveness of the system to the changing needs of each student over time through continued use of the system.

After developing a mature learning system through continued use, the proposed design plan builds a solid adaptive and intelligent learning system that combines AI, adaptive learning, learning analytics, and tailored feedback. Also, the systems support data-driven education practices that promote better learning, higher academic achievement, and improved learning experiences for individual students in the STEM program at Kapitolyo High School.

Artificial Intelligence Algorithms Used in the System

The Learning Management System (LMS) integrates Artificial Intelligence (AI) to provide adaptive learning pathways, personalized feedback, and real-time analysis of student knowledge. The system intentionally uses a hybrid AI approach composed of: **(1) Bayesian Knowledge Tracing (BKT)** for mastery estimation and adaptive learning progression, and **(2) Large Language Model (LLM)-based generative AI** for dynamic question creation and individualized feedback. These two components work together to personalize assessments and adjust difficulty levels according to each learner's performance and learning needs.

1. Bayesian Knowledge Tracing (BKT)

The Bayesian Knowledge Tracing (BKT) algorithm models how students acquire specific skills or concepts over time by estimating the probability that a learner has mastered a topic based on their observable performance (Corbett & Anderson, 1995). BKT assumes that learning is not directly visible and must be inferred from correct or incorrect responses. BKT treats each learning objective as binary (either known or unknown) and updates mastery based on four probabilistic parameters:

Table 2

Bayesian Knowledge Tracing (BKT) Variables Used

Symbol	Meaning
$P(L_0)$	The initial probability that the student knows the skill before any practice
$P(T)$	The probability that learning occurs after an opportunity to practice
$P(G)$	The probability that the student guesses correctly without knowing the skill

P(S)	Probability that the student slips (answers incorrectly despite knowing)
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Mathematical Model

If the learner's response is correct:

$$P(L_n|correct) = \frac{P(L_{n-1})(1 - P(S))}{P(L_{n-1})(1 - P(S)) + (1 - P(L_{n-1}))P(G)}$$

If the learner's response is incorrect:

$$P(L_n|incorrect) = \frac{P(L_{n-1})P(S)}{P(L_{n-1})P(S) + (1 - P(L_{n-1}))(1 - P(G))}$$

Accounting for learning after each observation:

$$P(L_n) = P(L_n|observation) + (1 - P(L_n|observation))P(T)$$

In the LMS, BKT processes quiz interactions, responses, and practice attempts to compute mastery levels in real time. If mastery is low, the LMS recommends remedial content or lowers the difficulty.

2. LLM AI Question & Feedback Engine (Llama 3.3-70B-Instruct)

The system integrates a generative Artificial Intelligence component powered by Llama 3.3-70B-Instruct Large Language Model, accessed through the Proxies API, to produce personalized learning materials and adaptive feedback. Unlike Bayesian Knowledge Tracing (BKT), which focuses on tracking mastery, the Large Language Model (LLM) serves as the system's content generation engine. It uses natural language generation to create questions, explanations, hints, and motivational prompts that are tailored to each learner's needs. When a student interacts with lessons or learning activities, the AI analyzes their responses in relation to their mastery status estimated by BKT. Based on this analysis, the Large Language Model (LLM) generates questions at appropriate difficulty levels, ensuring that learners are neither overwhelmed with advanced content nor bored with overly simple tasks.

Large Language Model (LLM) also produces detailed explanations after assessments, allowing students to understand why their answers were correct or incorrect. This supports meaningful learning by guiding students to reflect on their reasoning process rather than merely presenting the right answer. Additionally, the AI provides personalized recommendations on what lessons to review, what skills to practice, and how to improve performance. It can even offer motivational feedback that encourages persistence and engagement, especially in an adaptive learning environment. Through this integration, the system achieves a dynamic instructional process where generated questions, reinforcement strategies,

feedback, and lesson suggestions are continuously aligned with the learner's real-time performance profile.

Synergistic Interaction Between BKT and Large Language Model (LLM)

The adaptive capability of the system emerges from the complementary interaction between Bayesian Knowledge Tracing (BKT) and the Llama 3.3-70B-Instruct Large Language Model (LLM) generative engine. BKT functions as the system's cognitive monitoring mechanism by estimating a student's mastery level for each concept in real time. Every interaction within the LMS, such as answering a question or responding to a quiz item, updates the probability that a specific skill has been learned. These evolving mastery estimates serve as the foundation for personalized decision-making. Instead of merely categorizing learners as "high performers" or "low performers," BKT provides a nuanced and continuously updated understanding of what each student knows and how likely they are to succeed in subsequent tasks.

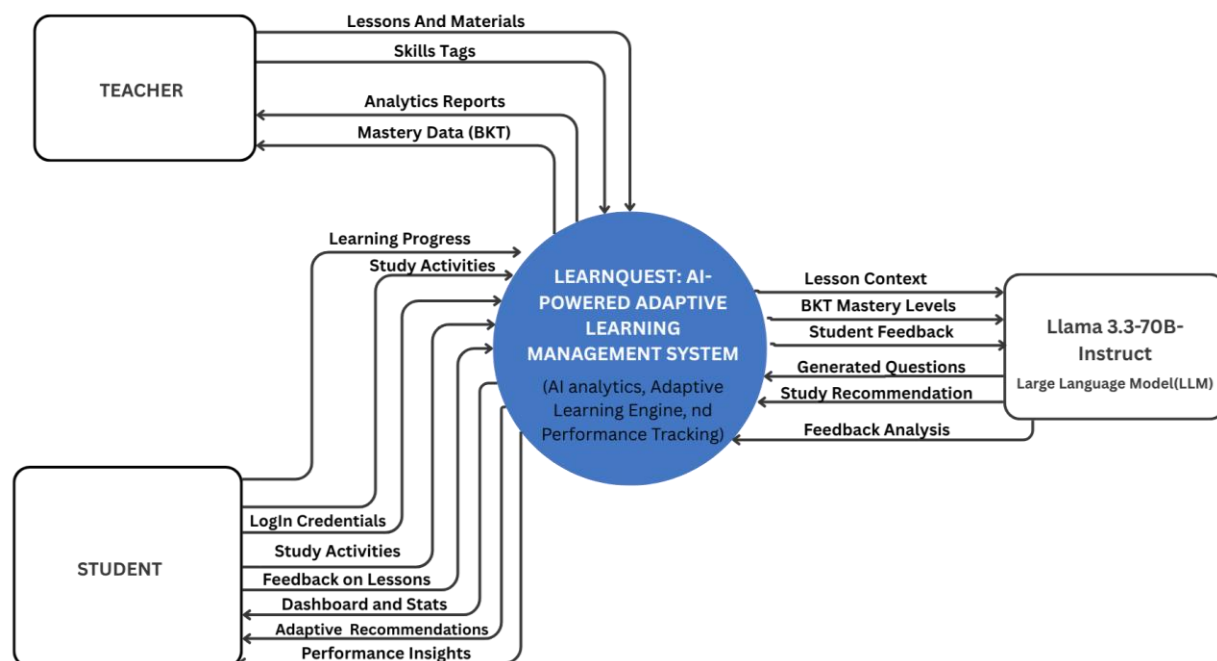
Large Language Model (LLM) uses these mastery insights to produce personalized learning experiences. When the system detects that a student has difficulty mastering a concept, the AI generates additional questions with simplified difficulty or provides scaffolded explanations that target the observed weaknesses. In contrast, when the mastery probability is high, Large Language Model (LLM) responds by creating more challenging questions or advanced learning tasks that

prevent stagnation and promote deeper understanding. This interplay ensures that students receive neither repetitive remedial content nor tasks that exceed their current learning capacity. In addition to adjusting difficulty, the AI explains errors, recommends specific lessons to revisit, and delivers motivational feedback that encourages learners to continue engaging with the material.

Together, BKT and Llama 3.3-70B-Instruct Large Language Model (LLM) establish a continuous feedback loop where performance data is analyzed, interpreted, and translated into actionable learning interventions. The student's responses inform BKT's mastery estimation, the mastery estimation guides Large Language Model (LLM) in generating personalized content, and the new content prompts further interaction that feeds back into the system. This closed cycle allows the LMS to remain responsive, personalized, and goal-oriented, enabling the development of an individualized learning pathway that supports the strengths and addresses the weaknesses of each learner. Through this synergy, the system moves beyond static instruction and becomes an adaptive environment capable of supporting the varied learning trajectories of senior high school STEM students.

Figure 2

Level 0 Data Flow Diagram of the System



Level 0 Data Flow Diagram

The Level 0 Data Flow Diagram for LearnQuest: An Adaptive Learning Management System, powered by Artificial Intelligence, for Kapitolyo High School's STEM Students, shows the flow of data between the system and its external entities, Students, Teachers, and Administrators, and the Llama 3.3-70B-Instruct Large Language Model (LLM). The diagram shows the flow of data for adaptive learning, AI-based assessment generation and recommendations, and mastery learning through Bayesian Knowledge Tracing (BKT).

The primary users of the system have been identified as Students. The primary activities that they perform are logging in, accessing learning resources, completing assessments, and submitting feedback. Once logged into the system, Students are taken to their dashboard, where they can view their progress towards

completing assessments, their performance, and the recommendations provided to them based on the responses and learning activities they provide to the system. The system captures the responses and learning activities of students; it tracks a student's level of mastery using BKT, and based on the student's level of mastery, the system provides the student with the appropriate level of assistance.

In the platform, teachers can create and manage instructional content. They also add lessons, instructional resources, and competency tags that structure the content to correspond with learning goals and objectives. Added content does not go unnoticed. Teachers gain access to analytics and mastery data generated from BKT. Reports give teachers insight into how students are performing and help them understand where students need support and where gaps in learning exist. System Administrators work on the backend and have a different set of responsibilities. They are concerned with the creation of user accounts, adjusting and maintaining the overall configuration and functionality of the system. They also have access to reports and statistics that aid them in assessing the system.

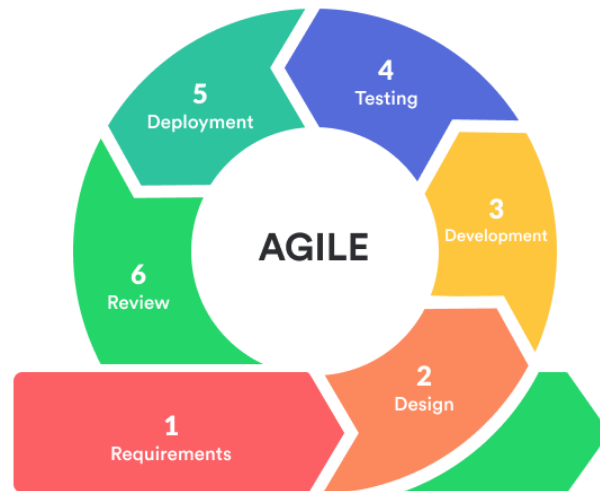
The Llama 3.3-70B-Instruct Large Language Model (LLM), an Artificial Intelligence component of the system, provides context for lessons through the learning management system, inputs from learners, and mastery data from BKT, for AIs to evaluate these data and generate feedback as to the recommended next steps in learning. This results in providing learning support for each learner based on their needs. In addition, the process facilitates teachers by providing instructional support that is properly targeted.

The LearnQuest system serves as the brain for the entire program. It collects and organizes data from users such as students, teachers, administrators, and AI to keep track of various learning materials, test scores, progress records, and skill assessments. The system always processes this data continuously to identify individual learners' progression, adjust to their needs, and provide appropriate suggestions. The organized, precise, and readily accessible data support the learner throughout the learning process.

At a high level, the Level 0 Data Flow Diagram shows how LearnQuest ties together users, course content, and AI tools into one adaptive learning space. As data flows back and forth and gets processed, the system can offer personalized learning, keep an eye on performance, and support decisions backed by real data. In the end, both students and teachers get helpful insights to boost learning. Plus, educators get clear, actionable info that can guide their teaching strategies and help improve student outcomes.

Figure 3

Agile Software Development Diagram



Software Engineering Paradigm

This study adopts the Agile Software Development Paradigm, particularly the iterative and incremental model, as the directing paradigm for the development and improvement of the LearnQuest: An Adaptive Learning Management System Powered by Artificial Intelligence for STEM Students of Kapitolyo High School.

The Agile method puts strong emphasis on flexibility, teamwork, feedback, and continuous improvement. This approach is different from traditional linear procedures such as the Waterfall method. It emphasizes user focus and flexibility. Agile can adapt based on stakeholder feedback and changing needs.

Agile development, as per Highsmith (2002), is such a development process that enables adaptation to change in a more efficient way, and it involves continuous interaction between developers, users, and all stakeholders. This approach focuses on developing functional software through iterations, during

which new features are added to the software. This is an excellent alternative for initiatives that use new technologies, including Artificial Intelligence (AI), which frequently require evaluation and improvement by getting people engaged.

In this research study, the use of the Agile method means that the teacher, student, and administrator are all involved in every stage of the LMS system development process. This involvement means that the LMS will remain relevant and effective in serving its learning purpose. The paradigm's repeat cycles let the researchers quickly find usability issues, change AI algorithms, and make learning better overall.

Agile Development Process

In this study, the agile methodology involves an iterative and incremental development approach consisting of multiple iterations of development. Each iteration results in the production of a prototype that is tested, improved upon, and then moved forward to the next phase. The different stages involved in the Agile development process have been explained below.

Planning Phase. In this phase, the researchers work together with teachers and learners to determine the requirements of the system, learning goals, and core functionalities of the LMS. The user stories are then collected and prioritized in order to develop a product backlog that will be used as a guide during each development iteration. These are some of the features identified: AI-generated

feedback, competency mastery tracking, adaptive assessments, and user access management.

Design and Prototyping Phase. Based on the requirements that have been gathered, plans are made for the system architecture and interface designs. Wireframes and data flow diagrams are made to show how users (teachers and students) would interact with the system. AI data pipelines are being planned to make the system adaptable and scalable.

Sprint Implementation: The Development Phase. During this phase, the development team will develop the system functionalities as per the planning done. The initial iterations will concentrate on the basic functionalities of the learning management system that include authentication, course management, uploading of learning materials, and quiz management. In subsequent stages, the other functionalities like integration of AI, Bayesian knowledge tracing, feedback mechanism, learning recommendation engine, and learning analytics will be included in the system. The developers validate the code's functionality repeatedly while they keep track of different versions.

Testing and Feedback Phase. During the completion of each sprint, the developed prototype is tested through the collaboration of the researchers and the selected teachers and learners. Some of the tests carried out include:

- **Functional Testing** – Ensuring the system does what it was supposed to do.

- **Usability Testing** – Checking the navigability, interface design, and user experience in general.
- **AI Output Testing** – Ensuring the quality, relevance, accuracy, and suitability of the outputs generated by AI, including summary, quiz questions, feedback, and recommendations, by evaluating these with teachers and subject matter experts.

During this phase, feedback collected from users and evaluators is noted down, analyzed, and incorporated into the further development process.

Review and Evaluation Phase. The team meets with the stakeholders to assess the completed iteration and address the issues raised, as well as figure out how to improve the process for the next iteration. Feedback from the stakeholders allows the team to make improvements to the system, optimize the AI-generated results, fine-tune the Bayesian Knowledge Tracing parameters, and optimize adaptive learning.

Deployment and Maintenance Phase. The LMS is ready for use in classrooms when the final version has been checked. The Agile methodology, on the other hand, stays in place after deployment. Maintenance and improvements are done as ongoing iterations. To make it last and work well in the long run, bugs, new features, and performance enhancements are added in later cycles.

The Advantages of Using Agile in this Study

The process of development of LearnQuest: an Artificial Intelligence-Powered Adaptive Learning Management System for Grade 11 and Grade 12 STEM Students of Kapitolyo High School, follows the Agile Software Development method as it is flexible, iterative, and learner-focused. The Agile process allows for the continual creation and improvement of the system via feedback from teachers, learners, and researchers throughout the development process.

As LearnQuest integrates artificial intelligence, adaptive learning, Bayesian knowledge tracing (BKT), and learning analytics, the agile methodology helps in implementing and testing each of these elements incrementally. With every iteration of software development, changes can be implemented considering user feedback to ensure that the content produced by the system, assessment capabilities, dashboarding, and recommendations provided serve the educational requirements of teachers and learners. Moreover, Agile methodology facilitates the quick identification of problems and their resolution to improve the system.

Retrieval-Augmented Generation (RAG)

To increase the accuracy of the content generated by AI, LearnQuest uses the Retrieval-Augmented Generation (RAG) strategy. This involves teachers feeding the system with their educational materials in the form of PDF files, Microsoft PowerPoint presentations, lectures, learning modules, worksheets, and lab manuals. Before generating educational content, the system extracts and

retrieves relevant information from these uploaded learning materials and provides it as contextual input to the Llama 3.3-70B-Instruct model.

The probability of AI hallucination, consistency of facts, and alignment of the lesson summary, quiz, and learning suggestions to the required science competency and lesson objectives is achieved by ensuring that the responses of the AI system are based on instructional materials from the teacher and not only the prior knowledge of the model.

AI Output Validation

The quality of the Artificial Intelligence component is evaluated through AI output validation. The generated lesson summaries, quizzes, personalized feedback, and learning recommendations are reviewed by science teachers to determine their educational quality and suitability before they are used by students.

The output generated is evaluated based on the following criteria:

- Accuracy of scientific concepts
- Alignment with the intended Science competencies
- Relevance to the uploaded learning materials
- Clarity and comprehensibility of generated content
- Educational appropriateness of assessment items and feedback

In case the outputs provided by artificial intelligence are wrong, confusing, and not consistent with the objectives of the lesson, teachers will have an option

to change, refuse, or create new material. Through this process, teachers ensure that artificial intelligence is just used as a means of a decision-making tool, not to replace the professional judgment of the teacher.

User Acceptance Testing (UAT)

The LearnQuest system will undergo User Acceptance Testing (UAT) to assess the usability and functionality of the system as well as the overall user satisfaction. A selected number of Grade 11 and 12 Science STEM students and Science teachers from Kapitolyo High School will participate for testing purposes.

These participants will test the LearnQuest system according to its ease of use, the effectiveness of AI-generated lesson summaries, the quality of the adaptive quiz, the importance of personalized feedback, and the clarity of learning recommendations and dashboards. The results obtained from the user acceptance testing will be used to find out the necessary changes to be made before the final deployment of the system.

Radar Charts for Learning Analytics

LearnQuest utilizes radar charts as part of its learning analytics dashboard to visualize student competency mastery across multiple Science learning competencies. Instead of presenting only overall assessment scores, radar charts display mastery levels estimated through Bayesian Knowledge Tracing (BKT),

enabling both teachers and students to identify strengths and areas requiring improvement.

In the form of radar charts, there is learning data presented multi-dimensionally, which consists of proficiency levels achieved, progress made, assessment performance, and learning gaps. Giving a visual depiction of learner performance, it aids in making evidence-based decisions regarding instructional decisions, personalized intervention, and adaptive learning pathways that are tailored to the needs of each learner.

Population and Sampling

This research will focus on the students and faculty members from Kapitolyo High School STEM academic Strands, who impacted the adaptive learning management system that is powered by artificial intelligence. The number of STEM students is 60 students in Grade 11, 50 students are in Grade 12, for a total of 110 students in the academic school year 2025-2026. These learners represent the primary intended users of the system and therefore serve as the primary focus of this study. The initial implementation of the LMS focuses specifically on their engagement with Chemistry subjects to ensure a controlled and discipline-specific analysis. Additionally, the teacher population consists of STEM faculty members, particularly those teaching Chemistry, who play a role in handling instructions and managing student participation within the LMS environment.

The participants of this study were selected using purposive sampling methods. This is a suitable sampling method because the system should be evaluated by users who have had direct interaction with the system and can provide meaningful observations about the usability and effectiveness of the system, as well as the overall learning outcomes. By sampling students who are specifically studying Chemistry, the data collected in this study will accurately reflect the true experience of users in the academic setting in which they participate.

A total of thirty-five (35) participants were selected through purposive sampling to participate in the system testing phase. These participants consisted of selected Grade 11 and Grade 12 STEM students who interacted directly with the LearnQuest system and provided meaningful feedback regarding its usability, effectiveness, and adaptive learning capabilities.

Data Collection Methods

The researchers employed a structured questionnaire and the LearnQuest Adaptive Learning Management System prototype to collect data for this study. The questionnaire was developed using Google Forms to facilitate efficient data gathering and response management. It consisted of close-ended questions measured using a five-point Likert scale. Two sections made up the instrument. The first section collected data on each student's learning/assessment

experiences within existing circumstances, and the second section evaluated the students' opinions regarding the LearnQuest system with respect to usability, effectiveness of assessment, learning support, and the overall user experience. The research advisor reviewed and confirmed the validity of each item on the questionnaire regarding its clarity and suitability for this research.

The sample of participants was purposely drawn from the Grade 11 and 12 STEM students at Kapitolyo High School. Respondents were provided with details about the study's purpose, confidentiality protection of responses, and the right to withdraw from participation at any time before taking part in this study. The participants received orientation regarding the LearnQuest system before submitting their complete questionnaire evaluating the system.

The LearnQuest prototype served as a secondary data collection instrument by recording system-generated learning analytics. These data included assessment scores, quiz completion records, question response accuracy, competency mastery levels, and AI-generated learning recommendations. The system also collected performance information used by the Bayesian Knowledge Tracing (BKT) model to estimate student mastery across identified learning competencies. In addition, the platform generated radar chart visualizations that summarized individual strengths and weaknesses based on assessment outcomes.

As part of the system evaluation process, participants were asked to access learning materials, complete AI-generated assessments, and review the feedback

and analytics provided by the system. The generated quiz results and mastery records were automatically stored within the database for subsequent analysis. Once collected, the information obtained through the questionnaire and data generated from the LMS were maintained in an organized manner and secured to be ready for analysis and interpretation.

Data analysis of the responses was completed using descriptive statistics, including counts, frequencies, or percentages, as well as computing means using weights for each response to the questions, weighted mean for use on the following scale: 3.50 - 5.00 = positive evaluation; 2.50 - 3.49 = neutral evaluation; and 1.00 - 2.49 = negative evaluation.

All collected data were treated with strict confidentiality and were used solely for academic and research purposes. The results of the analysis provided insights into the usability, effectiveness, and acceptability of the LearnQuest Adaptive Learning Management System, particularly in its ability to generate assessments, evaluate student performance, identify learning strengths and weaknesses, and support adaptive learning through Artificial Intelligence, providing insights into the effectiveness of AI-generated assessments, mastery tracking, learning analytics, and adaptive learning support.

Ethical Considerations

The researchers conducted this study at Kapitolyo High School while ensuring the protected rights and privacy of all participants throughout the research process. Appropriate ethical measures and intellectual property rights were observed to maintain confidentiality in data handling since the study involved the collection of student assessment data and learning modules.

Respondents were informed of the study and its purpose before beginning data collection. Participation was completely voluntary, with members of the study group being given a reasonable opportunity to contact researchers with questions before providing consent to participate in the study.

To protect participant privacy, only information necessary for the conduct of the study was collected. Personal identifiers such as student names were not included in the datasets used for analysis. Instead, unique codes were assigned to participants to ensure anonymity. Access to collected data was restricted to the researchers and the research adviser only.

The security of data is essential when developing and implementing a system. LearnQuest System security measures were an essential part of the design of this system. Security features to protect user information from being compromised include an authenticated login process and the use of encryption to store passwords. All access to the information found in the LearnQuest System was based on restricted user roles, allowing students to access only their individual

records while allowing teachers to view all records associated with the classes they had been assigned. Administrative access was limited to maintenance and operational purposes only. The approved researchers also adhered to the requirements of the Data Privacy Act of 2012, the Republic Act No. 10173.

Since the use of Artificial Intelligence (AI) in this system requires incorporating consideration, instructions were provided to teachers outlining their role as the main evaluators and decision-makers for assessment decisions. Teachers will use AI-generated assessments for reference only and will use their own professional judgement and classroom observations when evaluating AI-generated results. The researchers also acknowledged that AI-generated outputs may not be completely accurate immediately after deployment due to limited learner data; therefore, to address this concern, the system was tested before deployment, and user feedback was considered in improving its accuracy and reliability.

The research participants were informed that the purpose of the system was to help students learn and identify areas for improvement, rather than to monitor or punish students. Both assessment results, as well as system-generated analytics from the participants and their summative assessments or measurement results, were to be collected solely for educational and research purposes. All information gathered during the study was treated with strict confidentiality and was not disclosed to unauthorized individuals or organizations. Data retention and

disposal procedures were to follow the procedures outlined by institutional policy, and consent was given by all participants.

Before conducting the study, the researchers sought approval from their research adviser at Pamantasan ng Lungsod ng Pasig by making a request letter and securing permission from the administration of Kapitolyo High School.

Upon completion of the study, the researchers committed to providing Kapitolyo High School with a summary of the findings and recommendations. Documentation and reporting of issues that occurred during the system implementation were completed as needed by the research adviser and the school administration for purposes of accountability.

Summary

This study proposed the development of LearnQuest, an adaptive learning management system powered by artificial intelligence for STEM students at Kapitolyo High School. The study began in response to the challenges students face in traditional and existing online learning environments. These challenges include the limited availability of personalized learning support, adaptive assessment, and smart learning analytics. Though individuals using current LMS solutions have access to instructional materials and assessment tools, LMSs are generally unable to automatically identify a learner's strengths and weaknesses and provide tailored personalized learning support.

The review of related literature demonstrates how the use of learning management systems, artificial intelligence, auto-question generation, learning analytics, and adaptive technologies is becoming increasingly important to modern education. Prior research has also shown that the effective use of AI-based assessment systems, such as the use of Bayesian Knowledge Tracing and the use of learning analytics are effective method of tracking student development, measuring students' basic skill sets, and supporting personalized learning. Additionally, research has been conducted concerning automatic question generation and demonstrations of how AI can generate assessment items automatically, reducing the workload of teachers and providing an ongoing source of assessment.

To address these issues, the researchers proposed the LearnQuest, an artificial intelligence-based adaptable learning management system that has been created with an emphasis on STEM education for students in Grades 11 and 12 at Kapitolyo High School. The system enables teachers to upload educational materials, produce quizzes using large language models, and track student performance using integrated learning analytics. The system analyzes student assessment results using Bayesian Knowledge Tracing to estimate mastery levels across various competencies. It then offers personalized recommendations and visualizes learner strengths and weaknesses through radar chart dashboards.

Based on the Agile Software Development Model, this research project was designed through a developmental research design methodology. The project was

built around several iterations of planning, designing, developing, testing, and evaluating to create a functional learning solution to address identified learning challenges. The researchers collected data through administering structured questionnaires and system-generated analytics to students who interacted with the LearnQuest prototype. Additionally, researchers performed descriptive statistical analyses on the data collected from the students to determine the system's usability, effectiveness, functionality, and user satisfaction.

The proposed system aims to enhance Science Technology Engineering Math (STEM) education by providing students with adaptive assessment tools, artificial intelligence (AI) based learning analytics, and data-driven instructional resources. By combining AI technology with features such as automatic question generation, use of Bayesian Knowledge Tracing (BKT) methods, and recommendations for personalized learning, LearnQuest intends to enable both teachers and students to make informed decisions regarding their learning, achieve a higher level of academic performance, and improve their overall learning experiences at Kapitolyo High School.

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